

Sri Lanka Institute of Information Technology



Assignment 02 (Individual) – GAMES DEVELOPMENT

SE4031 – Games Development

Title – 2D Interactive Game

Assignment Created By: Mr. Aruna Ishara Gamage, Mr. Nushkan Nismi

Weight: 20%

Arcane Academy: Skyward Broom

IT22006202 – K A B Sandeepa



Summary of the Game Concept

"Arcane Academy: Skyward Broom" is a 2D interactive fantasy survival game. The player controls a brave young witch navigating through a dangerous forbidden sky corridor above the Magic Academy, riding her trusted broomstick. The core objective is to survive as long as possible while the sky view automatically scrolls forward, representing her flight through the grounds. The player must dodge floating gargoyle statues and arcane anomalies (obstacles), banish dangerous dark entities like spectral bats and hovering Dementors (enemies) using powerful spell bolts, and collect restorative elixirs (collectibles) to stay alive. The game gets progressively harder as the auto-scrolling flight speed increases over time.

How the Game Works

- **Health:** You start with 100 health, which is shown on the screen. Hitting an obstacle drops your health by 10, and crashing into an enemy drops it by 20. If you hit an obstacle, you take damage but fly right through it. If your health hits 0, the game stops. When you restart, your health goes back to 100.
- **Score:** Your score is displayed on the screen. You earn points by staying alive longer and by shooting enemies. Whenever you restart the game, your score resets to 0.
- **Healing Potions:** Floating magic books are easy to tell apart from enemies. Grab them to instantly fill your health back up to 100. If you fly past them, they safely disappear off-screen to keep the game running smoothly.
- **Magic Attacks:** Left-click your mouse to shoot magic bubbles. The spell shoots straight ahead from your witch and disappears if it hits an enemy or travels too far.
- **Enemies:** Dark spirits randomly appear in front of you. If you crash into them, you lose health. If you shoot them with a magic bubbles, they disappear, and you get extra points!
- **Increasing Difficulty:** The game slowly speeds up as you play, making it trickier to survive the longer you fly.

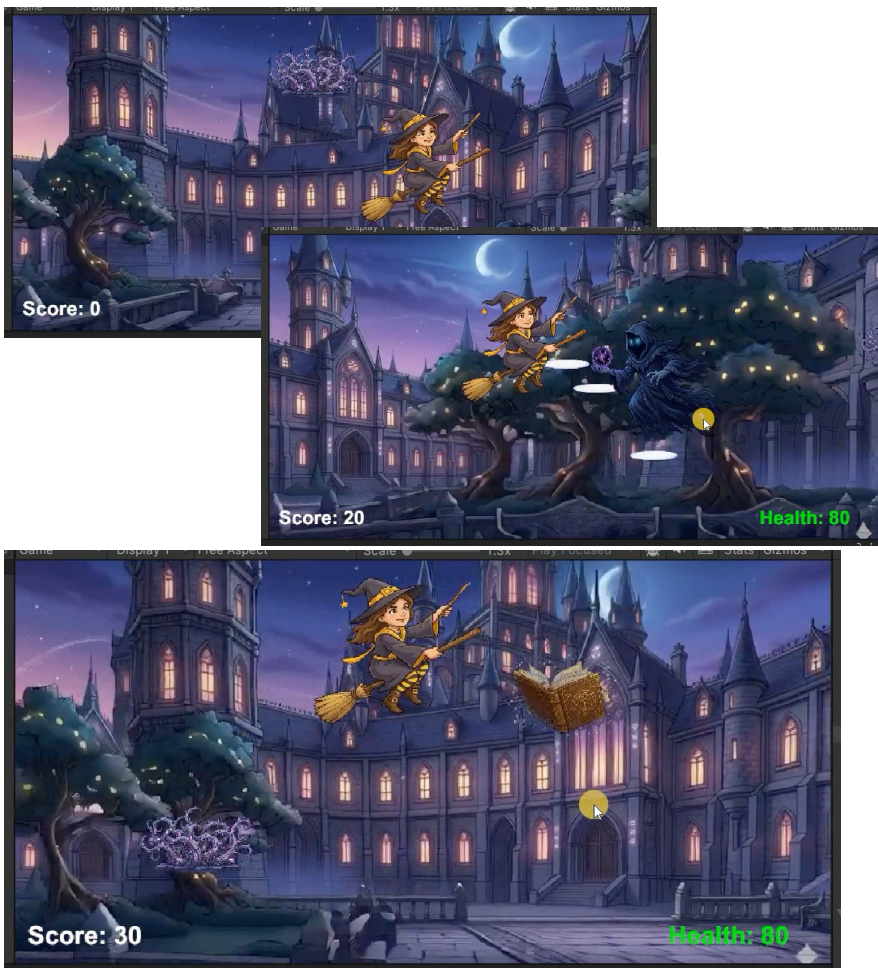
Control Guide

- **W / Up Arrow:** Move Up
- **S / Down Arrow:** Move Down
- **A / Left Arrow:** Move Backward
- **D / Right Arrow:** Move Forward
- **Left Mouse Click:** Cast Spells (Projectile)

Creative Feature

Dynamic UI Health Indicator & Custom Audio System As a creative modification, the game's Health UI constantly updates its color based on the player witch's current health status (Green for High Health, Yellow for Medium, and Red for Critical). Furthermore, a custom audio manager was implemented with individual volume controls for different events, featuring unique sound effects for taking damage from a gargoyles obstacle vs. a dark entity, consuming a restorative elixir, and casting a spell, layered over a continuous magical background ambiance and track.

Screenshots of Gameplay



Enemy



HealthPack



Bubbles



Credits and External Assets

- **Game Engine:** Unity (2022.3)
- **Visual Assets (Sprites & Background):** Generated by Gemini AI and Meta AI.
- **Original Game Logic & Author:** IT22006202 – K A B Sandeepa
- **Audio Assets (Music & SFX):** Downloaded from Pixabay (Royalty-Free Sound Effects).

Gameplay Demo Video Link

https://drive.google.com/file/d/1sdtTPwrike_WEMiOZ3ffNEUxVTIc_WWk/view?usp=drive_link